

# Project Initialization and Planning Phase

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| Date          | 29 September 2026                                                   |
| Team ID       | SWTID-2026-6605                                                     |
| Project Name  | Script-Controlled ACL – Restrict Record Access Based on Field Value |
| Maximum Marks | 5 Marks                                                             |

## Project Proposal (Proposed Solution) report

The proposed solution secures the ServiceNow Institution Details table using role-based and script-controlled ACLs. It addresses unauthorized record visibility and operation access by evaluating user roles together with the Branch value of the current record.

|                   |                                                                                                                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Overview  |                                                                                                                                                                                                                                         |
| Objective         | Implement a secure ServiceNow ACL model that restricts record access by role and Branch value while allowing administrators full access.                                                                                                |
| Scope             | Create the Institution Details table, users and roles, then configure and test Read, Create, Write and Delete ACLs.                                                                                                                     |
| Problem Statement | Users with different responsibilities require different levels of access. Unrestricted access can expose EEE, ECE or CSE records to unauthorized users.                                                                                 |
| Impact            | A controlled ACL model reduces unauthorized viewing, creation, modification and deletion of Institution Details records and makes permissions easier to test and manage.                                                                |
| Proposed Solution | Use script-controlled Read access for EEE records and separate role-based ACLs for Create, Write and Delete operations.                                                                                                                 |
| Approach          | Create roles bb1, bb2, bb3 and bb4. The Read ACL checks admin status, bb1 role and current Branch = EEE. Additional ACLs use bb2 for Create, bb3 for Write and bb4 for Delete.                                                          |
| Key Features      | <ul style="list-style-type: none"><li>• Role-based authorization</li><li>• Record-level Branch restriction</li><li>• Administrator full access</li><li>• Separate CRUD permissions</li><li>• Impersonation-based verification</li></ul> |

## ACL Permission Model

| Operation   | Role  | Access Rule                                                                                               |
|-------------|-------|-----------------------------------------------------------------------------------------------------------|
| Read        | bb1   | Can read only records where Branch = EEE. Administrator is allowed full access.                           |
| Create      | bb2   | Allows creation of Institution Details records when the required role is present.                         |
| Write       | bb3   | Allows editing of records when the required role is present; Read ACL still governs record visibility.    |
| Delete      | bb4   | Allows deletion when the required role is present; Read/Write security remains relevant to record access. |
| Full Access | admin | Administrator retains unrestricted access to the table.                                                   |

## Resource Requirements

| Resource Type | Description              | Specification / Allocation                     |
|---------------|--------------------------|------------------------------------------------|
| Platform      | Application platform     | ServiceNow instance                            |
| Data          | Custom record storage    | u_institution_details table                    |
| Security      | Roles and ACLs           | admin, bb1, bb2, bb3, bb4                      |
| Testing       | User access verification | Impersonation of authorized/unauthorized users |